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MOSAIC: Mobile Object Segmentation Under Adverse Imaging Conditions For Rapid L-PBF Keyhole Behavior Characterization

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In the laser powder bed fusion (L-PBF) additive manufacturing process, the rapid evolution of gas and the molten metal interactions complicates proper monitoring and control, with unstable keyholes leading to porosity and spatter formation. A deeper understanding of these interactions will improve the ability of process engineers to develop control schemes for the L-PBF process. Operando synchrotron X-ray imaging of the L-PBF process has provided ground truths for keyhole dynamics by revealing the phenomena occurring beneath the material surface at high frame rates. Previous work requiring quantifiable segmentations has used manual image labeling or specifically trained machine learning models, but a robust, standardized method remains needed for this task, which is critical for on-site experimental troubleshooting or for processing large, diverse imaging datasets. In this work, we propose a standardized pipeline for characterizing the morphological behavior of the keyhole under varying imaging conditions and image quality, achieving an average F1 score of 0.893 and an accuracy and precision of 0.98 and 0.953, respectively.

MOSAIC, a Mobile Object Segmentation algorithm for experiments under Adverse Imaging Conditions, is designed to perform rapid, accurate analysis of keyhole dynamics during active X-ray imaging experiments without requiring time-consuming manual labeling or model training. We validate the performance of MOSAIC by analyzing 12 X-ray imaging datasets and comparing its segmentation against labels generated from an aggregate of manual annotations. Each dataset has a unique combination of experimental parameters, such as substrate composition, laser power, scan speed, shutter speed, and frame rate, to ensure the method's wide applicability. Two examples of segmented images are shown in Figure 1. We also test the performance of the developed algorithm by sweeping the values of selected processing and segmentation parameters, demonstrating that the segmentation output of MOSAIC is tractable and deterministic. The results from a parameter sweep study show that normalizing the temporal window used to calculate the background image by the frame rate, velocity, and pixel size improves performance by ensuring proper separation between the keyhole and its surroundings. MOSAIC's optimal temporal window occurs when the normalized temporal window exceeds the moving object's pixel width, thus demonstrating the algorithm's tractability.

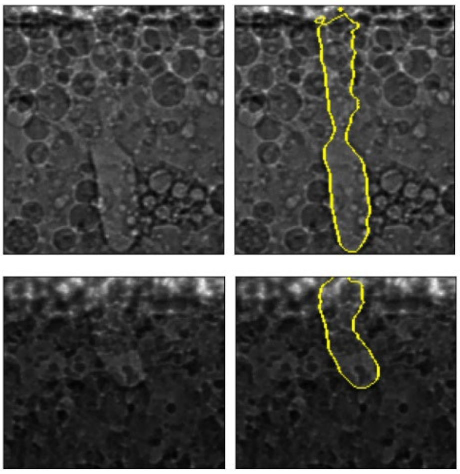

Figure 1. Keyhole images that have been segmented for an Al-(top) and Ti-(bottom) alloy as-built substrate with powder present.

The open-source algorithm will enable those running high-speed operando X-ray imaging experiments on the L-PBF process to perform real-time analysis and ensure accurate collection of the desired information. The discoveries made via high-speed operando X-ray imaging experiments have been critical in understanding the L-PBF process, and the proposed MOSAIC algorithm will assist future researchers in making new discoveries. In future work, MOSAIC's deterministic segmentations can serve as a deployable tool for agentic AI. Specifically, we will integrate agentic AI with the MOSAIC method to link complex physics observed phenomena in imaging data to in situ process monitoring data, opening a new frontier for improving L-PBF performance through monitoring and control systems.